Power Tips for Intel-based Systems

The right installation will pave the way for a trouble-free operation and let you harness the full potential of your PC. Keep in mind the following tips while buying and installing an Intel processor.

Keeping your cool: Since processors are continuously run at high speeds, cooling becomes an important issue—gift your processor a high rated heatsink combined with the best cooling fan.

Power to run: Due to additional power requirements, all Pentium 4 processors need the extra +12 V rail to run correctly. Therefore, make sure the power supply for your new processor has the necessary four-pin +12 V connector that goes into the ATX12V connector on your motherboard.

Go for quality: Don't pinch pennies and choose a cheap motherboard. A high-quality motherboard may cost a few thousand rupees more, but it will guarantee far greater stability and support for the hardware, the chipset and the BIOS. You'll also get more features, such as more overclocking options and integrated peripherals.

BIOS updates: Make sure you buy a motherboard that offers regular BIOS and driver updates on the company Web site. Regular BIOS updates means great support for newer hardware, and helps iron out any bugs and flaws. Newer BIOS updates can also mean significant performance boosts.

To HyperThread or not?: Intel's new 3.06 GHz processor now incorporates a technology called HyperThreading, which enables the processor to perform far better when running multiple applications simultaneously. However, this comes at a price, and you should opt for it only if you need high performance for executing multiple tasks. You will also need to confirm if your motherboard supports HyperThreading.

A question of memory: The current scenario gives users a choice of SDRAM, DDR and RDRAM memory in a Pentium 4-based system—DDR memory offers high data transfer bandwidth and is priced lower than RDRAM. However, RDRAM's higher data transfer rate makes it ideally suited to demanding applications such as gaming, CAD modelling and graphics-intensive, texture-heavy applications. Also, RDRAM needs to be installed in pairs, unlike DDR or SDRAM—this needs to be kept in mind while buying memory modules. These three types of memories cannot be used on the same system, so choose with care as you will not be able to change your memory type without changing the motherboard, if you need to do so in the future.

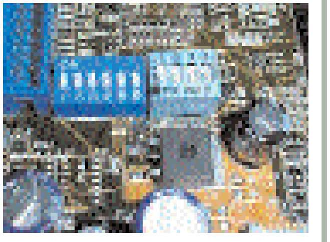
Willamette or Northwood?: While the newer Northwood Pentium 4 processors—built around the 0.13-micron fabrication process—are more popular than their older 0.18-micron Willamette counterparts, the latter is still available in the market. Willamette processors draw more power and therefore generate more heat. Also, Northwood processors integrate twice the amount of L2 cache (512 KB) as compared to 256 KB integrated by the Willamettes.

Special drivers: Intel supplies drivers for the new 800 series chipsets (including the 845, 845-DDR and 850 chipsets) called the Intel Application Accelerator driver. These increase the performance of your IDE devices significantly. They offer better boot times and also surpass the 137 GB limitation of the IDE drives in the native Windows drivers.

Application support for special instruction sets: Intel processors have always had special instruction sets such as SSE2 that help deliver better performance with certain types of data. You may need to install specific patches that let these applications use the special instruction sets.

Workshop: The right way to install a processor

STEP 1. Preparing the motherboard

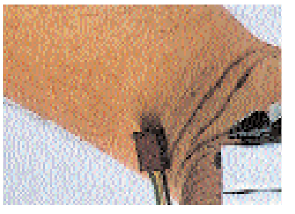


Set your motherboard to run at the correct core voltage, bus frequency and clock multiplier setting by consulting the manual. In some cases, you may need to do this through the DIP switches on the motherboard or the Frequency/Voltage settings in the BIOS.

Step 2. Preparing the processor

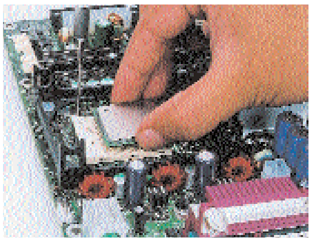
Apply thermal paste on your processor by using a thin plastic visiting card—a plastic card is best suited to spreading the paste smoothly across the surface of the processor core. You can also extend this paste around the core, but smoothen out the paste so that there are no air bubbles. Also, the paste should be in contact with every portion of the processor core.

Step 3. Preparing the heatsink



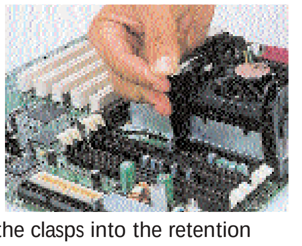
Locate the region on the underside of the heatsink where the processor is going to touch the processor core. This is usually in the centre of the flat area of the heatsink. Once you have ascertained this zone, wipe it clean of any dust. Then evenly spread some thermal paste, carefully avoiding any air bubbles.

Step 4. Seat the processor in the socket



Clamp the processor firmly into the socket on the motherboard and see that it is correctly aligned with the pins. Do so without forcing the processor.

Step 5. Install the heatsink



Place the heatsink on the processor gently, but firmly—and take care not to damage the core of the processor while doing this. Make sure that the heatsink is seated such that it is in complete contact with the processor core. After it has been positioned, fasten the clasps into the retention hooks of the processor socket to ensure that it is in firm contact with the processor.

Step 6. Connect the CPU fan

Finally, connect the CPU fan on the correct point on the motherboard. Make sure that the heatsink assembly is firmly seated—there should be no play in it. You're now ready to zoom!